Applied Machine Learning for Cybersecurity

Course Project Proposal

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Kairo: Attack Classification Using ML and DL Techniques

1. Understanding the Attack Problem

1.1. Overview of the HELLO_FLOOD Attack

In the context of RPL-based IoT networks, the *HELLO_FLOOD attack* is a type of **control plane flooding attack**. It targets the *routing infrastructure* of an RPL network by overwhelming nodes with a massive number of *HELLO (DIO)* messages. These messages are control packets typically used for establishing and maintaining network topology (DODAG).

1.2. Working Mechanism

In normal operations:

- DIO packets are periodically transmitted by nodes to share routing information such as **rank**, **preferred parent**, and **version**.
- Nodes process these packets to determine their position in the DODAG and maintain optimal routes.

In HELLO_FLOOD:

- A malicious node sends **continuous and excessive DIO packets**.
- Neighboring nodes become overwhelmed with fake control messages.
- This leads to **energy exhaustion**, **memory buffer overflow**, and **routing table corruption**.
- It may eventually cause **Denial of Service (DoS)** by exhausting the victim node's resources or delaying legitimate routing processes.

1.3. Attack Objectives

- **Disrupt network communication** by keeping nodes engaged in processing irrelevant control packets.
- **Prevent formation of correct routing trees (DODAG)**.
- **Degrade network performance** by increasing delay, energy consumption, and packet drop rate.
- **Reduce network availability** and impact mission-critical IoT applications like healthcare, smart homes, or industrial monitoring.

1.4. Impact on RPL Networks

Effect	Description
High packet traffic	Continuous unsolicited DIO messages flood the network.
Unstable routing paths	Frequent changes in parent nodes due to fake updates.
Node exhaustion	Energy and memory depletion in low-power IoT devices.
Increased delays	Queuing and processing time rise due to packet overflow.
Reduced throughput	Legitimate data gets delayed or dropped due to resource unavailability.

1.5. Why It's Challenging to Detect

- The attack *mimics legitimate behavior* (sending DIO packets), making it harder to identify through simple rules.
- Standard IDS may not handle *burst-based behaviors* without high false positives.
- Requires *temporal and behavioral analysis* of packet frequency and traffic patterns.

1.6. Relevance to DETONAR

The DETONAR paper targets such attacks using:

- Traffic sniffing
- Feature extraction (like DIO frequency, packet rates)
- Hybrid anomaly + signature-based detection

Our project will build upon these principles, using ML/DL to detect abnormal packet behaviors indicating HELLO_FLOOD activity.

2. HELLO_FLOOD Dataset Feature Analysis

Below is a detailed explanation of each column in the dataset, grouped into relevant categories based on their role in the network stack and potential usefulness for detecting a HELLO_FLOOD attack.

2.1. Control & Routing Metadata

Feature	Description	Relevance to Attack
PACKET_TYPE	Type of packet (Control_Packet / Application_Packet)	**High** – Filter DIO control packets.
CONTROL_PACKET_TYPE/ APP_NAME	Specifies DIO, DIS, DAO, etc.	**High** – DIO count is central to HELLO_FLOOD.
RPL_RANK	Node's logical position in DODAG	**Medium** – May help confirm manipulation.
RPL_VERSION	Version of DODAG instance	**Low** – Less useful for HELLO_FLOOD but relevant in version-based attacks.
PCKT_LABEL	Ground truth label (0/1) indicating attack or benign	**Essential** – Required for model training.

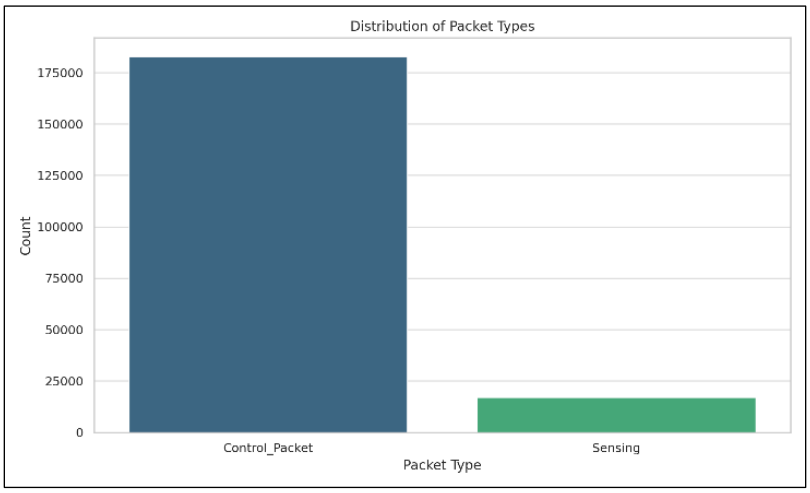


Figure 1: *Distribution of Packet Types in the RPL-based IoT Simulation.*

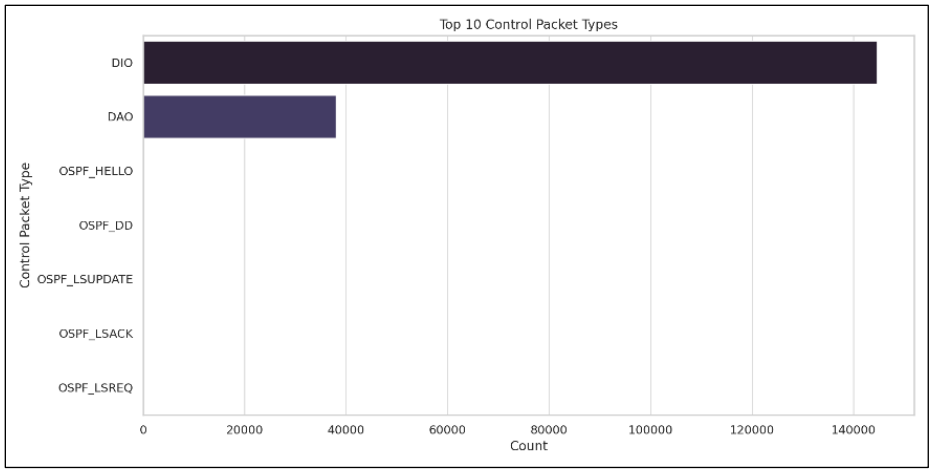


Figure 2: *Frequency of Top 10 Control Packet Types in the Simulation Dataset.*

2.2. Transmission & Node Metadata

Feature	Description	Relevance to Attack
SOURCE_ID / DESTINATION_ID	Logical IDs of the communicating nodes	**High** – Helps identify attacker and traffic flow.
TRANSMITTER_ID / RECEIVER_ID	Nodes actually sending/receiving packet	**High** – More physical view of transmission.
SOURCE_IP / DESTINATION_IP	IP address mappings	**Medium** – Correlates with IDs if needed.
GATEWAY_IP / NEXT_HOP_IP	Routing path info	**Medium** – Could help identify abnormal routing behavior.

2.3. Time Features (Per Layer Arrival/Start/End Times)

Feature	Description	Relevance to Attack
PHY_LAYER_START/END/ ARRIVAL_TIME(US)	Physical layer timestamps	**High** – Time series trend of packets per node.
MAC/NW/APP/ TRX_LAYER_ARRIVAL_TIME(US)	Packet arrival times per stack layer	**Medium** – Not all used; redundancy possible.

2.4. Payload Size Features

Feature	Description	Relevance to Attack
APP/TRX/NW/MAC/ PHY_LAYER_PAYLOAD(Bytes)	Payload size at each layer	**Low** – HELLO_FLOOD packets tend to be small and uniform.
PHY_LAYER_OVERHEAD(Bytes)	Protocol overhead	**Low** – Doesn't vary much during attacks.

2.5. TCP/Transport-Like Fields (Mostly Null)

Columns (all NULL): SEGMENT_ID, SEGMENT_LEN, SEQ_NO, ACK_NO, isSyn, isAck, isFin, CWND, LOCAL_ADDRESS, REMOTE_ADDRESS, Unnamed: 38
These fields are irrelevant for HELLO_FLOOD, as it is a control-plane routing attack and does not involve TCP sessions. They can be safely dropped.

3. Feature Selection for HELLO_FLOOD Detection

3.1. Selecting and Justifying Important Features

Based on our dataset analysis and the DETONAR methodology (which relies on packet sniffing and temporal patterns), the following features are identified as critical for detecting HELLO_FLOOD attacks:

Feature Name	Justification
PACKET_TYPE	To filter out control packets (Control_Packet), since HELLO_FLOOD specifically targets DIO transmissions.
CONTROL_PACKET_TYPE/APP_NAME	Identifies DIO messages; attack detection depends on excessive DIO frequency.
SOURCE_ID / TRANSMITTER_ID	Helps identify which node is sending the suspicious number of control packets. TRANSMITTER_ID gives physical behavior; SOURCE_ID gives logical node identity.
RECEIVER_ID	Complements transmitter-side analysis and allows understanding of traffic distribution.
PHY_LAYER_START_TIME(US)	Key to time series analysis—detects packet bursts over time. Used in ARIMA modeling in DETONAR.
PHY_LAYER_END_TIME(US)	Same role as above. Could be used to estimate transmission duration or density.
PCKT_LABEL	Required for supervised ML models as it indicates the presence of attack (1) or benign traffic (0).
NEXT_HOP_IP / GATEWAY_IP	Helps map network topology and detect abnormal routing behaviors that might arise from flooding.
RPL_RANK	While not essential, this may assist in validating that the attack does not trigger unintended topology shifts, supporting anomaly detection logic.

Why do the above selected features work?

- DETONAR's hybrid detection relies on *temporal frequency* of DIO packets → timing & control type needed.
- Source/transmitter IDs are required to *identify the attacker*.
- The label is crucial for training/evaluating detection models.

3.2. Dropping Irrelevant or Redundant Features

Some features do not contribute meaningfully to HELLO_FLOOD detection or are mostly null. These can be dropped for the following reasons:

Dropped Feature	Justification
TRX_LAYER_ARRIVAL_TIME, APP_LAYER_ARRIVAL_TIME, NW_LAYER_ARRIVAL_TIME	Redundant with PHY/MAC timing and often null.
SEGMENT_ID, SEGMENT_LEN, SEQ_NO, ACK_NO, isSyn, isAck, isFin, CWND	TCP-level details. HELLO_FLOOD is not a transport-layer attack.
LOCAL_ADDRESS / REMOTE_ADDRESS / Unnamed: 38	All-null or placeholder columns. Not useful.
Payload size features (APP/MAC/NW/TRX/PHY)	Packet sizes for HELLO packets are generally uniform and do not exhibit distinguishing patterns.
RPL_VERSION	Only has 1 unique value in the dataset, making it non-informative for learning.

4. Feature Relationships in HELLO_FLOOD Detection

To detect a HELLO_FLOOD attack, understanding how the selected features interact — both **logically** and **temporally** — is essential.

4.1. Temporal Relationships

The HELLO_FLOOD attack relies on *packet frequency and timing*, so *temporal features* are at the core.

- **PHY_LAYER_START_TIME(US)** and **PHY_LAYER_END_TIME(US)**:
  - Help calculate *packet rate* for each node over time.
  - Can be used to *window traffic* (e.g., every 10s or 30s) and compute the number of DIO packets sent by each node.
  - Anomalies such as *spikes in packet count* or *short inter-packet intervals* may indicate flooding.

These features form the **time series backbone** of the model. If a node sends significantly more DIO packets in shorter intervals, it can be flagged.

4.2. Node Behavior & Identity Relationships

These identifiers form a logical mapping of "who is doing what" in the network.

Feature	Purpose
SOURCE_ID	Logical identity of the source of the packet.
TRANSMITTER_ID	Physical node that transmits the packet.
RECEIVER_ID	Destination node that receives the packet.
NEXT_HOP_IP	Routing path from the node.
GATEWAY_IP	Gateway used for data forwarding.

Key Interactions between features:

- **SOURCE_ID ↔ PHY_LAYER_START_TIME**: Helps associate traffic rate with specific nodes.
- **SOURCE_ID ↔ CONTROL_PACKET_TYPE**: Filters per-node DIO control traffic.
- **NEXT_HOP_IP** might remain static for benign behavior but **shift erratically** if the topology is disrupted due to attack.

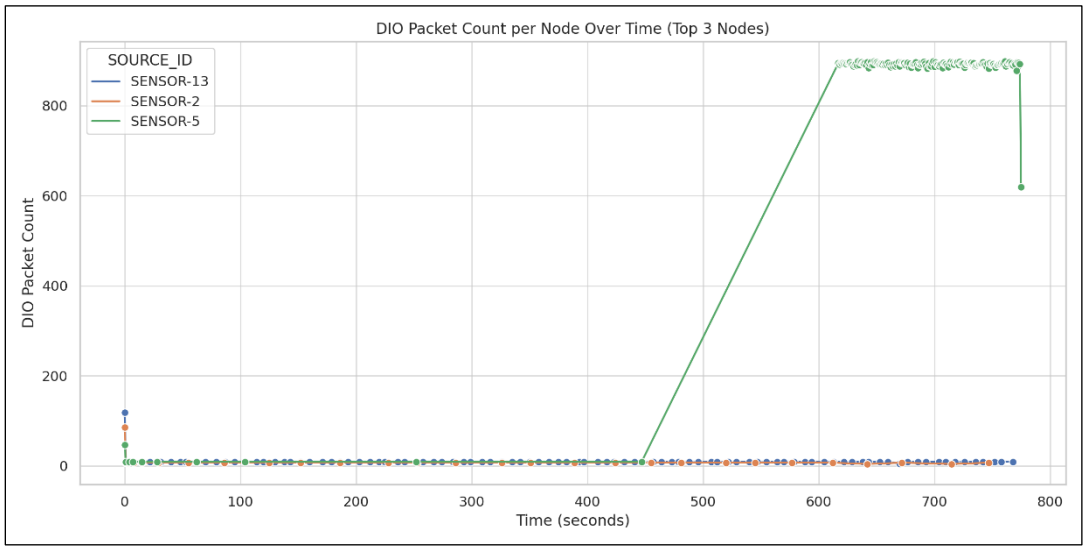


Figure 3: *DIO Packet Transmission Over Time for Top 3 Nodes. Peaks may indicate malicious flooding.*

4.3. Packet Content Relationships

Feature	Role
PACKET_TYPE	Ensures only Control_Packet is considered.
CONTROL_PACKET_TYPE/APP_NAME	Differentiates between DIO, DAO, etc. DIO is crucial.
RPL_RANK	Adds context: is a node pretending to be closer to the root?

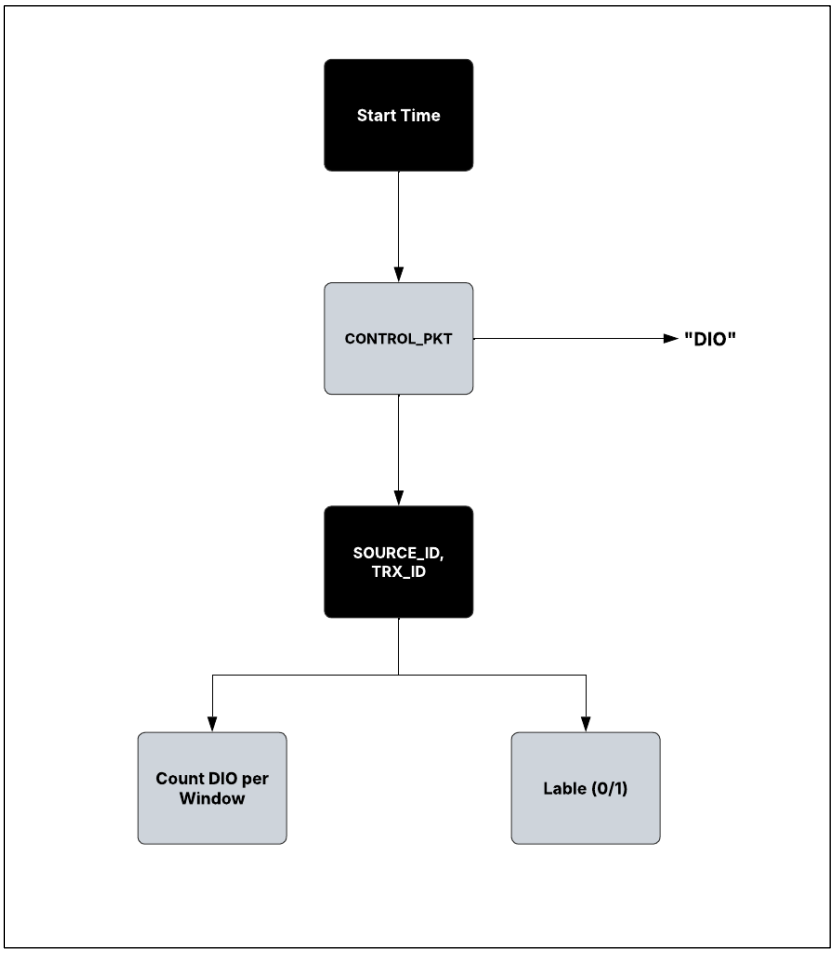
These help with *packet filtering* and *anomaly classification*:

- DIO flooding will show *many DIO packets* from a *single or few nodes*.
- RPL_RANK may reveal whether the node is pretending to be close to the DODAG root to attract traffic (relevant to sinkhole attacks but can support HELLO_FLOOD logic as well).

4.4. Ground Truth Connection

- **PCKT_LABEL**: This is the **label column** indicating whether a row belongs to a HELLO_FLOOD attack (1) or benign (0). It is essential for supervised ML model training and validation.

4.5. Conceptual Feature Relationship Flow



The selected features combine into a **behavioral time series** that allows:

- **Trend-based detection** (via ARIMA or moving averages),
- **Node-specific analysis** (who is flooding),
- **Label-based training** (via PCKT_LABEL).

This structured linkage paves the way for effective detection using time-series ML/DL models.

5. Proposed Models for Classification

After understanding the attack, exploring the dataset, selecting relevant features, and analyzing their relationships, we now propose suitable **Machine Learning (ML)** and **Deep Learning (DL)** models for effective detection of HELLO_FLOOD attacks in RPL-based IoT networks. We aim to:

- Detect **abnormal surges** in DIO control packets,
- Link them to specific nodes (attackers),
- While maintaining low false positives and fast detection time.

5.1. ARIMA (AutoRegressive Integrated Moving Average)

- **Type:** Time Series Forecasting (Unsupervised)
- Used in **DETONAR** for anomaly detection by forecasting normal packet behavior over time.
- Can predict future values of DIO counts per node and raise an alert if actual values exceed a threshold.

Benefits	Drawbacks
Lightweight and interpretable.	Requires manual tuning (p, d, q values).
Requires no label data (unsupervised).	**Per-node model** requirement makes it less scalable.
Effective in small-scale, static networks like in RADAR simulations.	Cannot classify attack types or adapt well to dynamic topologies.

5.2. Logistic Regression

- **Type:** Supervised Binary Classification
- Fits well when working with tabular, feature-engineered data (like packet count per window).
- Predicts PCKT_LABEL (0 = benign, 1 = attack).

Benefits	Drawbacks
Simple and interpretable.	Assumes linear relationship; may underperform with non-linear attack patterns.
Good baseline model for understanding feature impact.	Doesn't capture time-based dependencies.

5.3. Random Forest Classifier

- **Type:** Supervised Ensemble Model
- Captures non-linear patterns and interactions between features.
- Handles categorical and continuous variables.

Benefits	Drawbacks
High accuracy, resistant to overfitting.	Not suitable for real-time detection (batch processing).
Provides feature importance scores for interpretability.	Requires substantial memory in large deployments.
Handles outliers and noise well.	

5.4. MLP (Multi-layer Perceptron / Neural Network)

- **Type:** Supervised Deep Learning Model
- Can learn complex non-linear patterns between packet counts, time gaps, & source behavior.

Benefits	Drawbacks
Flexible architecture; can model intricate relationships.	Requires larger training data.
Performs well with balanced and structured features.	Black-box nature (less interpretable).
Handles outliers and noise well.	Needs tuning and regularization to prevent overfitting.

5.5. LSTM (Long Short-Term Memory Network)

- **Type:** Recurrent Neural Network for Time Series
- Ideal for **sequential data** like packet counts over time per node.
- Can detect **burst patterns** and **temporal anomalies**.

Benefits	Drawbacks
Captures long-term dependencies.	Requires larger datasets (long time series per node).
Effective for HELLO_FLOOD where timing is the main signal.	**Training is computationally expensive**.
	Needs input to be reshaped as a windowed sequence.

6. Model Comparison Table

Model	Type	Supervised	Captures Time?	Pros	Cons
ARIMA	Statistical	**No**	**Yes**	Lightweight, unsupervised	Not scalable per-node
Logistic Regression	ML	**Yes**	**No**	Simple, interpretable	Limited to linear separability
Random Forest	ML	**Yes**	**No**	Non-linear, robust	Batch-only, high memory
MLP	DL	**Yes**	**No**	Captures complex relations	Needs tuning, opaque
LSTM	DL	**Yes**	**Yes**	Best for time-based anomalies	Heavy training, harder setup

7. Strategy

Deployment Scenario	Recommended Model(s)
Lightweight, fast setup	ARIMA + Rule-based filtering
Initial ML baseline	Logistic Regression, Random Forest
Best DL-based detection	LSTM (with packet count sequences)
Balanced real-world use	MLP with sliding windowed features

8. Conclusion

- **ARIMA** is a solid choice for fast, rule-assisted detection like DETONAR.
- For ML-based classification using extracted features, **Random Forest** provides robustness and explainability.
- For a time-aware, attack-predictive model, **LSTM** is the most powerful — especially when generalizing to other routing attacks.